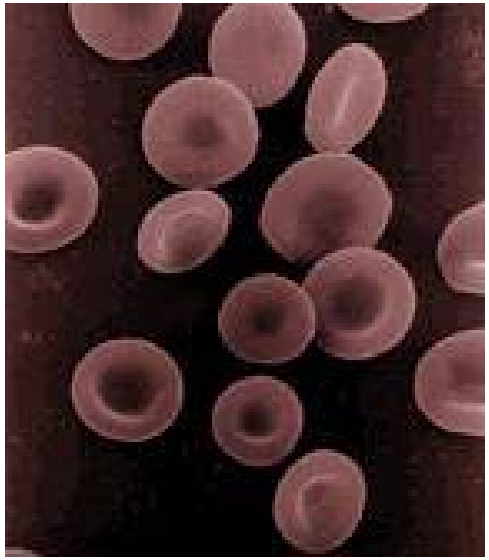
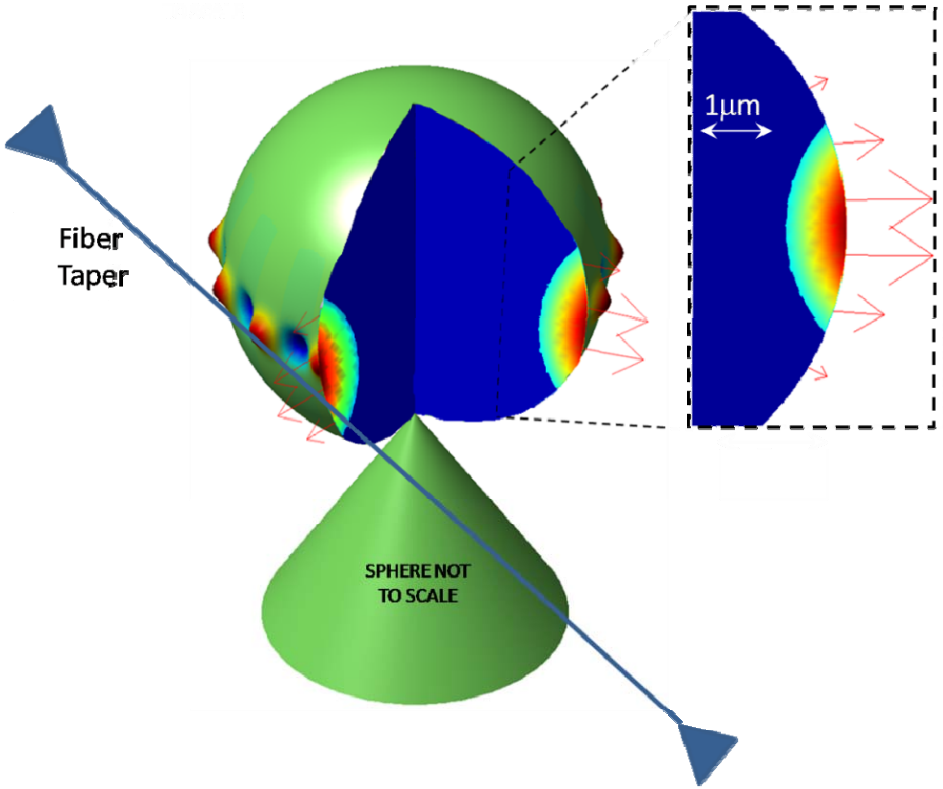
Surface Optomechanics: Mechanical Whispering Gallery Modes in Microspheres

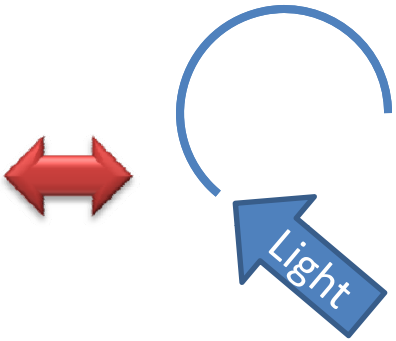




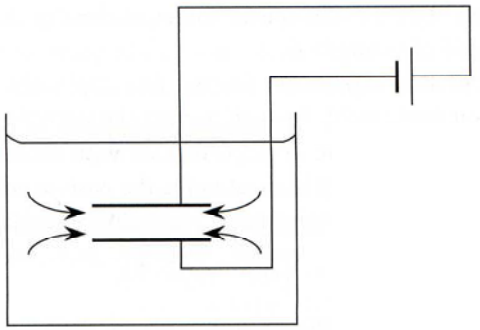
John Zehnpfennig, Matthew Tomes, Tal Carmon
The University of Michigan, Ann Arbor, USA

Various types of Optical Forces

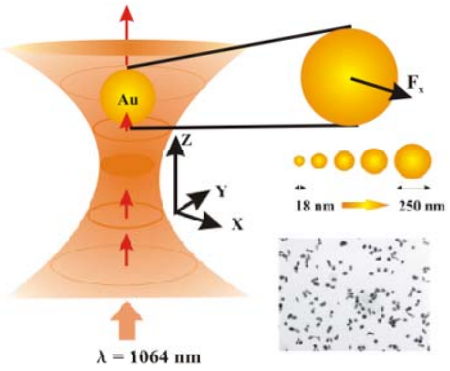




Scattering Force



Today, Compressive Force



Gradient Force

Poster Session, Gyroscopic Force

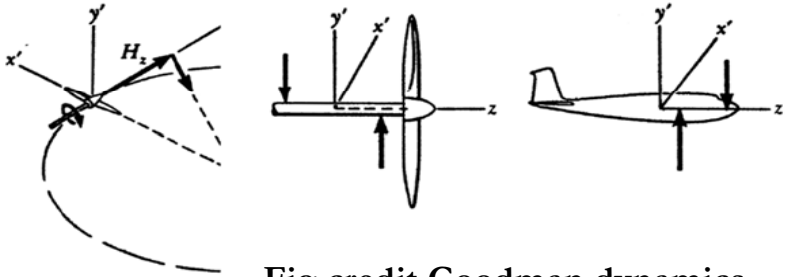
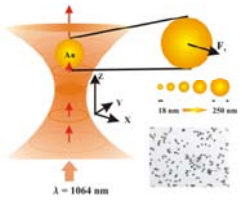
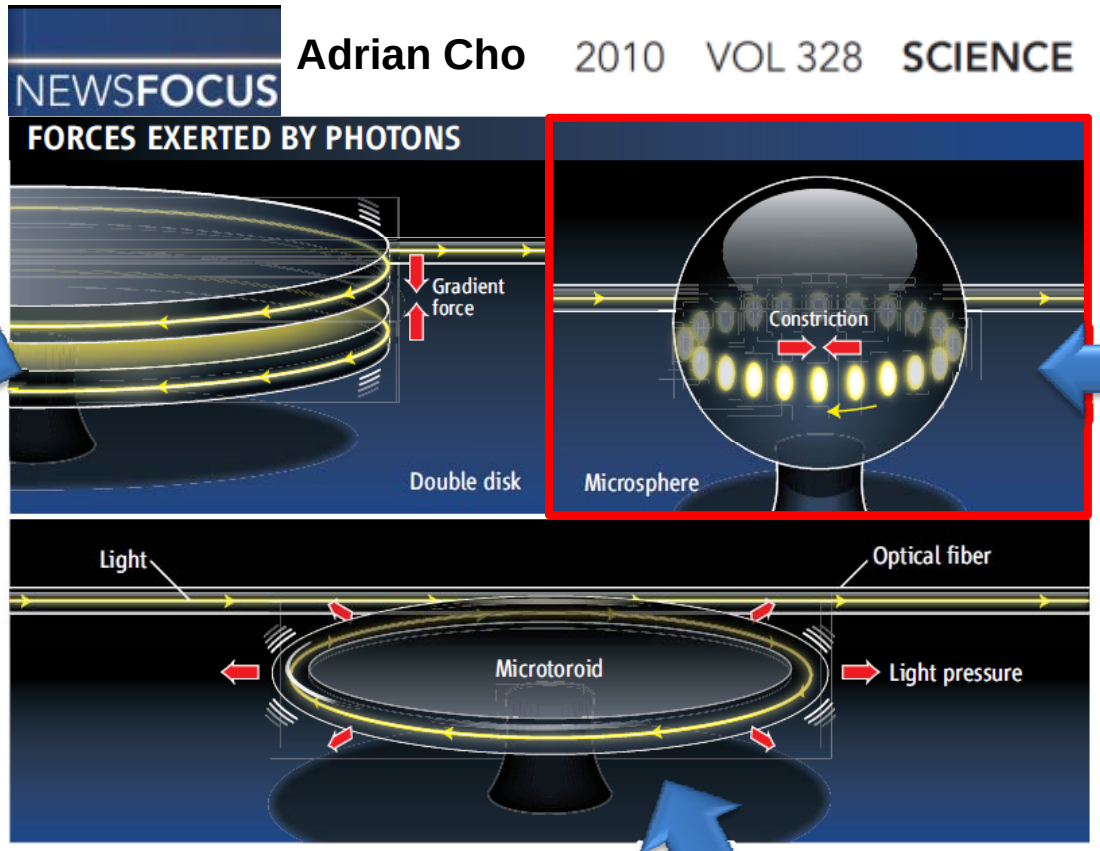
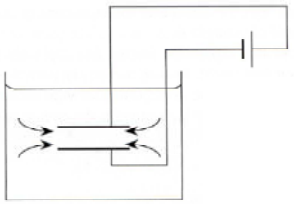


Fig credit Goodman dynamics
Zhang et. al. , 2010 (In Review)



Qiang Lin, et al.,
PRL, 2009.





Tomes, et al.,
PRL, 2009.



Carmon, et al.,
PRL, 2005.

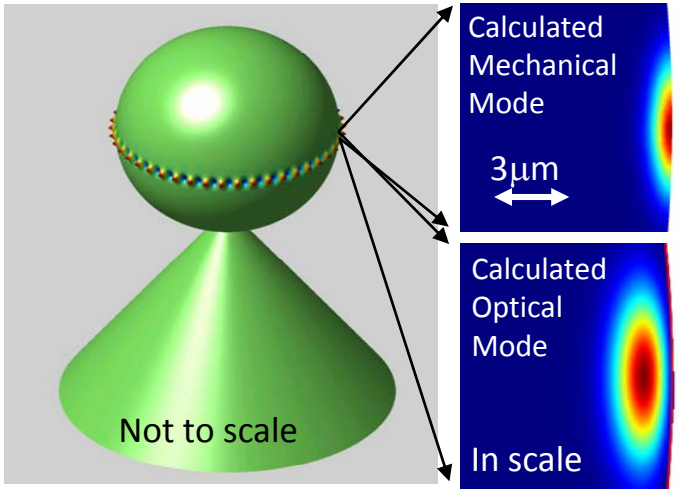
Figure Credit: Adrian Cho, Science 2009-2010 (2009).

WGMs, schematic description

Whispering gallery resonators

- Lord Rayleigh – "The problem of the whispering gallery" (1910)
- Light is trapped by total internal reflection
- The circumference is an integer number of wavelengths





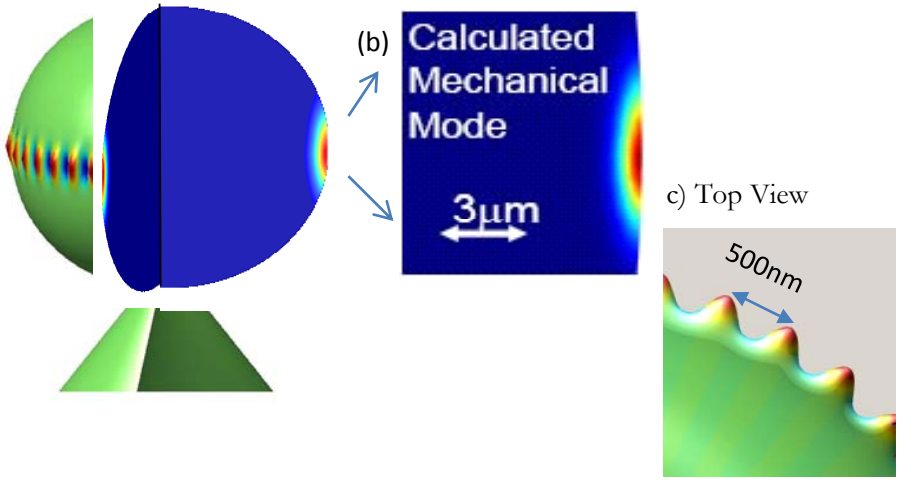
	Perturbation	Speed	Governing Eq.
Mechanical	Deformation	of Sound	Stress-Strain Eq
Optical	EM	of Light	Maxwell's Eq

WGMs, numerical solution

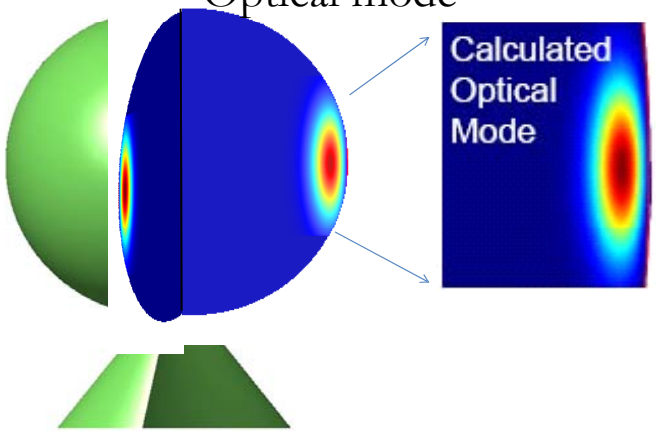
Solving for mode in whispering galleries

- Mechanical mode solution
  - Stress strain tensorial equations
  - Exact equations except for discreatization in space
- Optical mode solution
  - Fully vectorial solver for Helmholtz equation

Mechanical mode, deformation

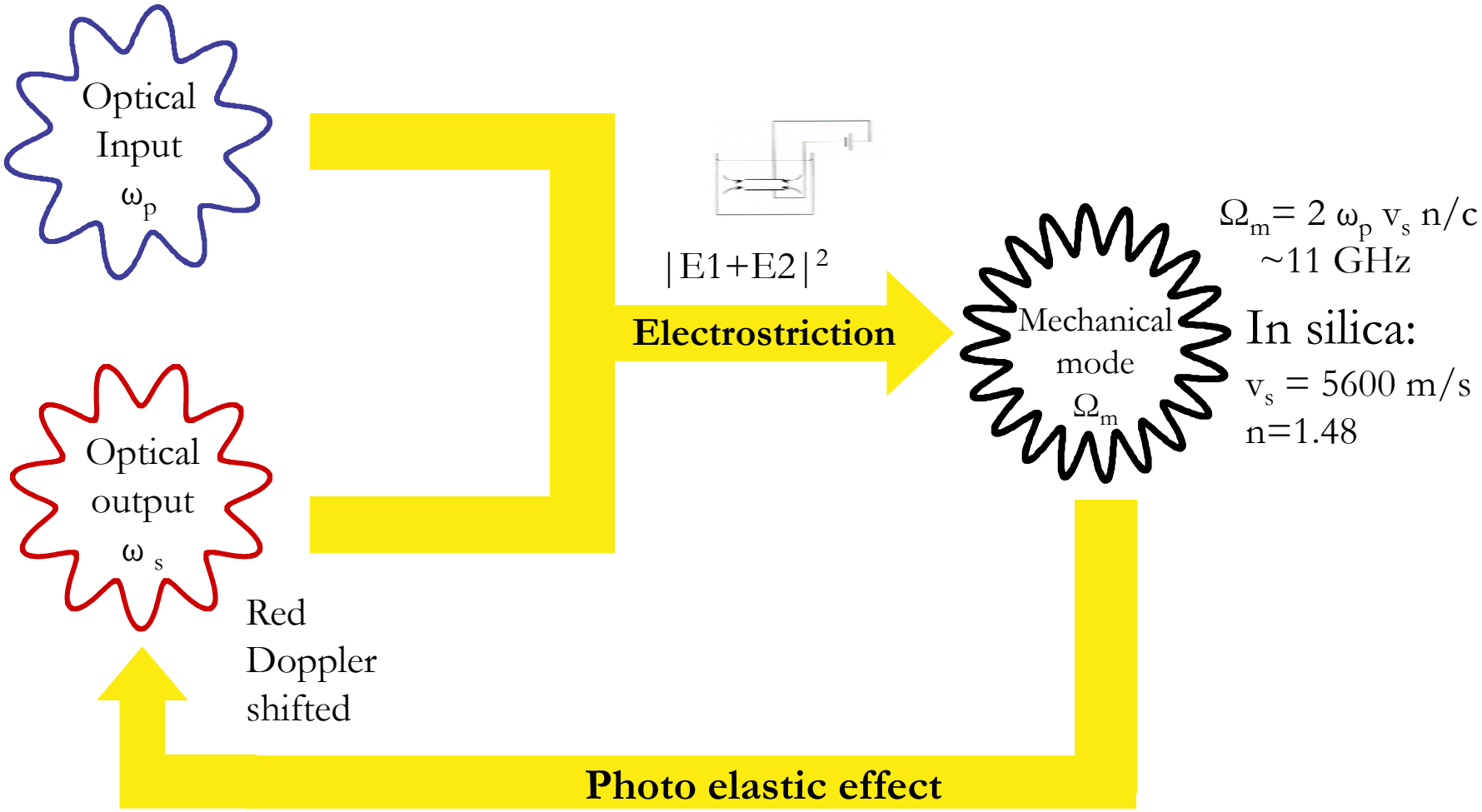


Optical mode



M. Oxborrow, IEEE Trans. Microwave Theory Tech. 55, 1209 (2007).

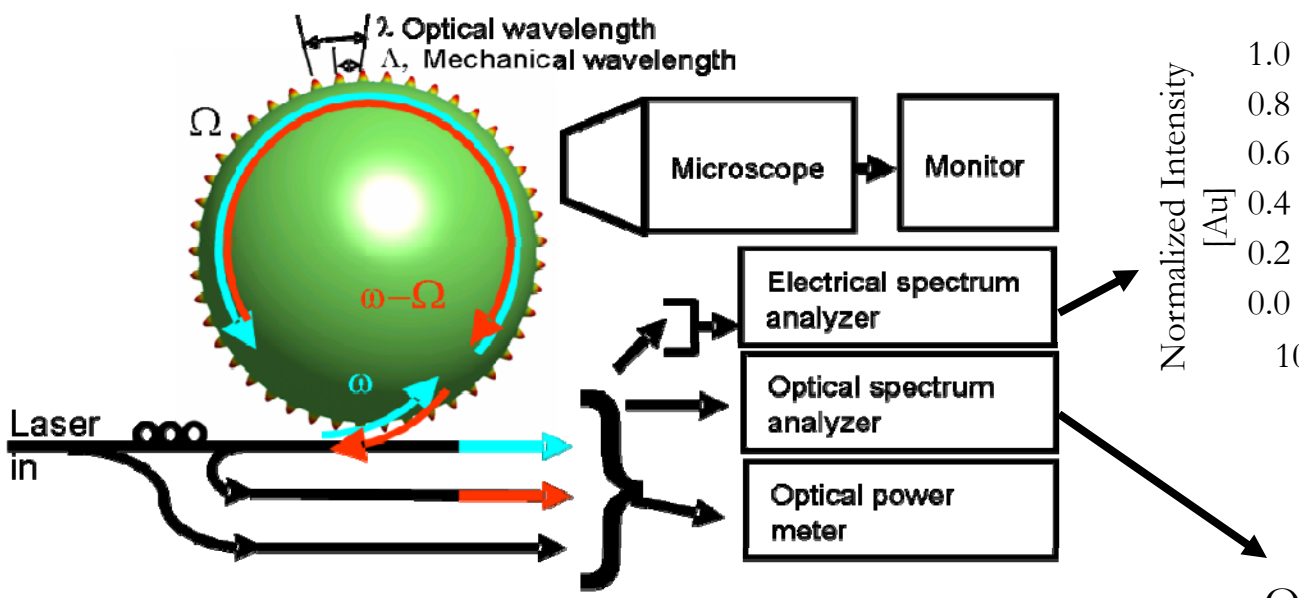
SBS Mode, Self Consistency

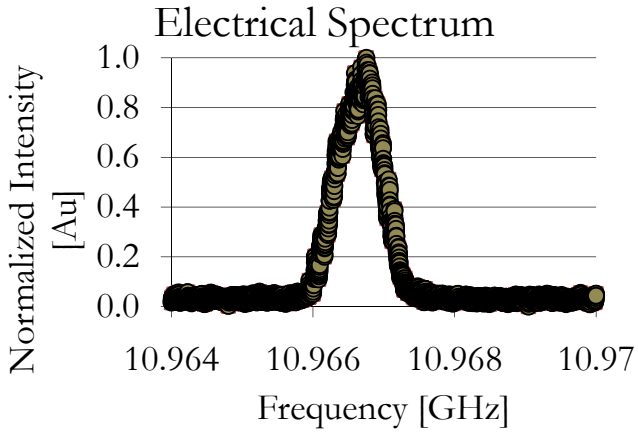


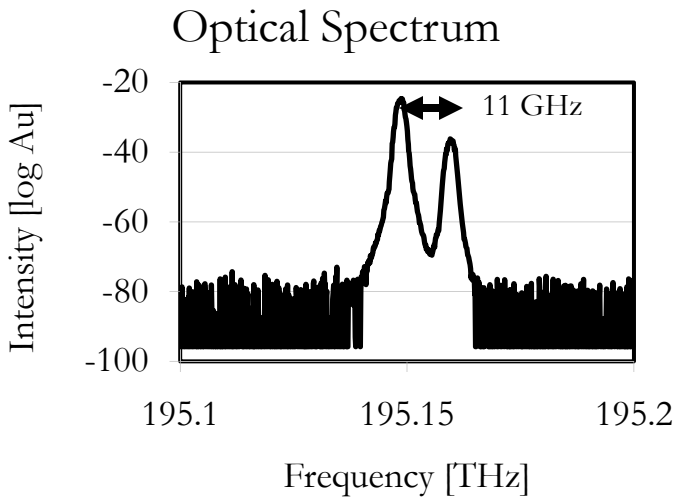
Effective propagation is 100 meters in a micron scaled device

A. Yariv, Quantum Electronics (Wiley, New York, 1975).

Experimental results: Vibration





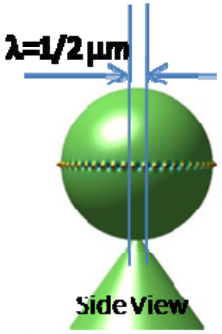


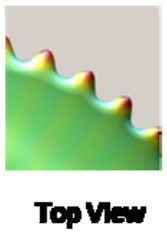
- Main claim:
  - 11 GHz Mechanical vibration rates

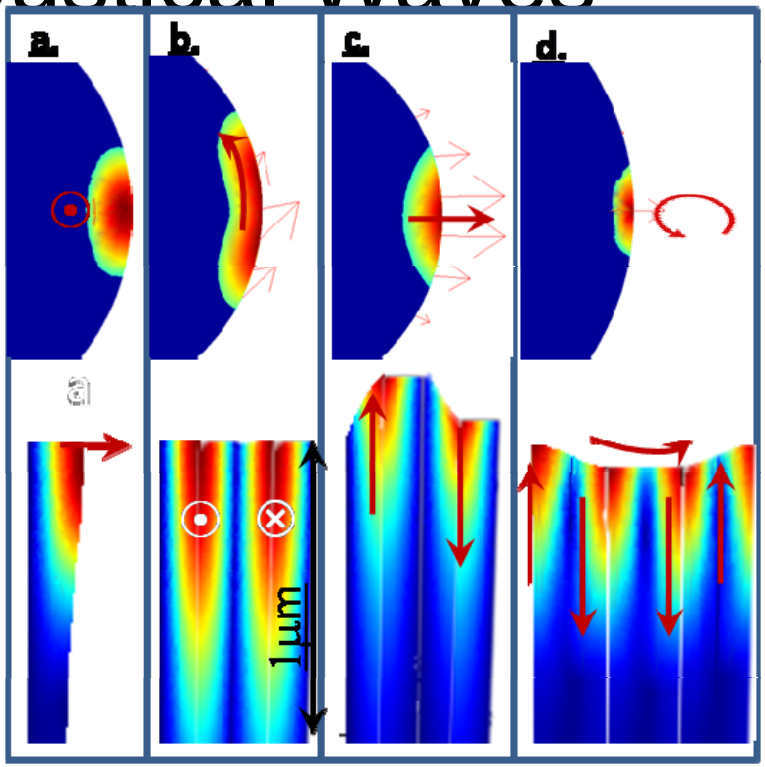
Matthew Tomes and Tal Carmon, Physical Review Letters **102**, 20-23 (2009).
J. C. Knight, G. Cheung, F. Jacques, and T. A. Birks, Opt. Lett. 22, 1129 (1997).
S.M. Spillane, T. J. Kippenberg, O. J. Painter, and K. J. Vahala, Phys. Rev. Lett. 91, 043902 (2003).
M. Cai, and K. Vahala, "Highly efficient hybrid fiber taper coupled microsphere laser," Opt. Lett. 26(12), 884– 886 (2001).

Different Types of Acoustical Waves

- There are several types of acoustical waves
- Here we look at
  - Longitudinal
  - Transverse Polar
  - Transverse Radial
  - Rayleigh
- Propagation velocities are different while wavelength is fixed



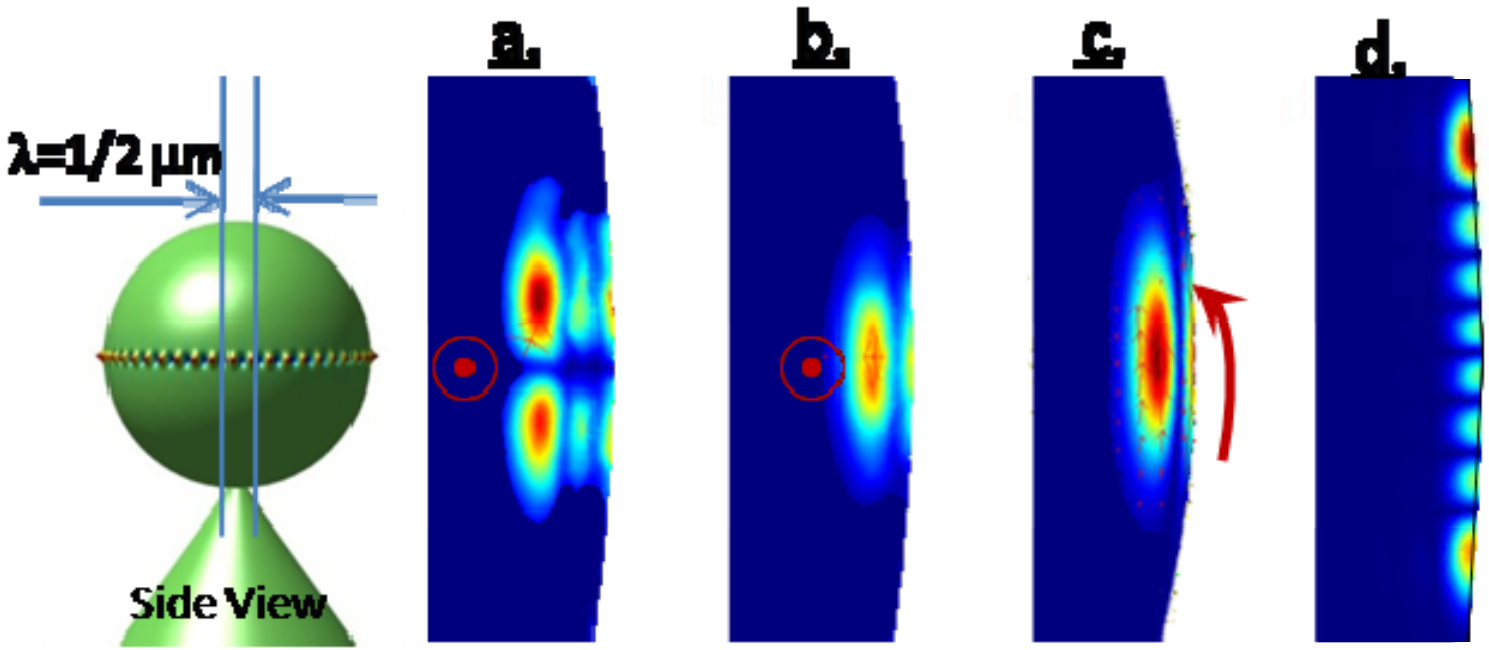




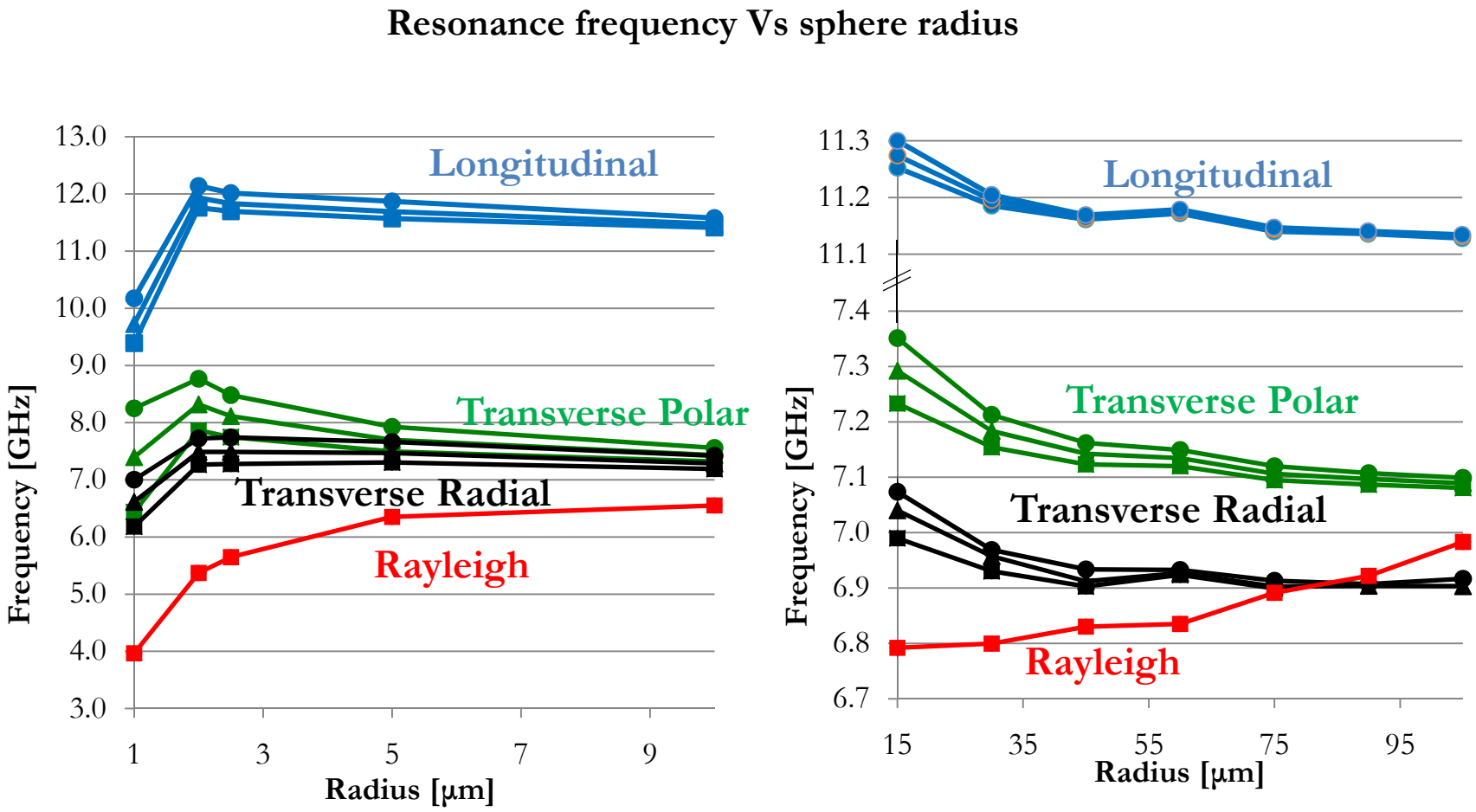
Bulk		60μm SiO_2 Sphere		
Wave	**Velocity [m/s]**	**Mode**	**Deformation**	**Velocity [m/s]**
Longitudinal	$V_L = \sqrt{\frac{E}{\rho}}\sqrt{\frac{(1-\nu)}{(2\nu^2+\nu-1)}} = 5848$	*Longitudinal*	East-West	5987
Shear	$V_S = \sqrt{\frac{E}{\rho}}\sqrt{\frac{1}{2(\nu+1)}} = 3687$	*Shear Polar*	North-South	3815
		Shear Radial	In-Out	3710
Rayleigh	$V_R = \sqrt{\frac{E}{\rho}}\sqrt{\frac{1}{2(\nu+1)}}\frac{(0.87+1.12\nu)}{(1+\nu)} = 3400$	*Rayleigh*	In-out and East-West	3663

High Radial and Polar Mode Order

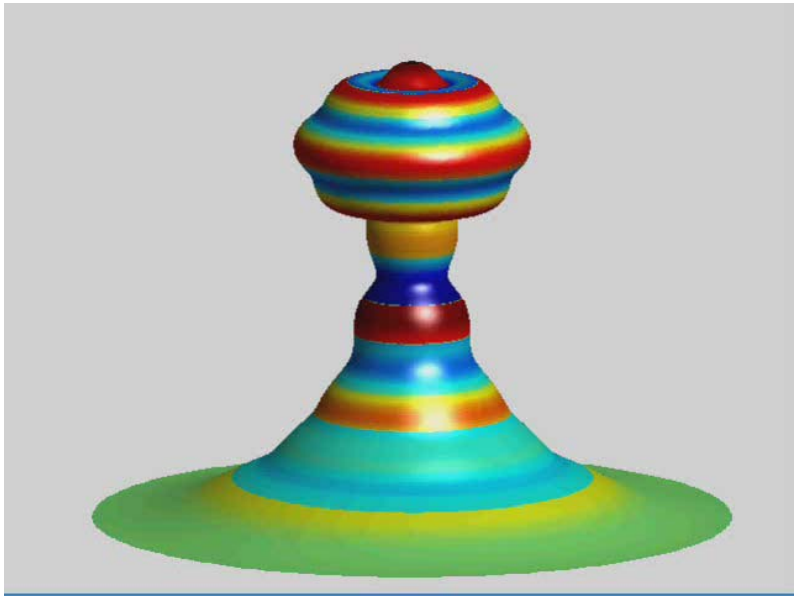
- For a 60 micron sphere, there are ~200 acoustical wavelengths along the circumference
- Transverse and polar directions can also have multiple maxima
- Different order modes can vary in frequency as a function of sphere size



Mechanical Frequency



Comparison with Others resonators

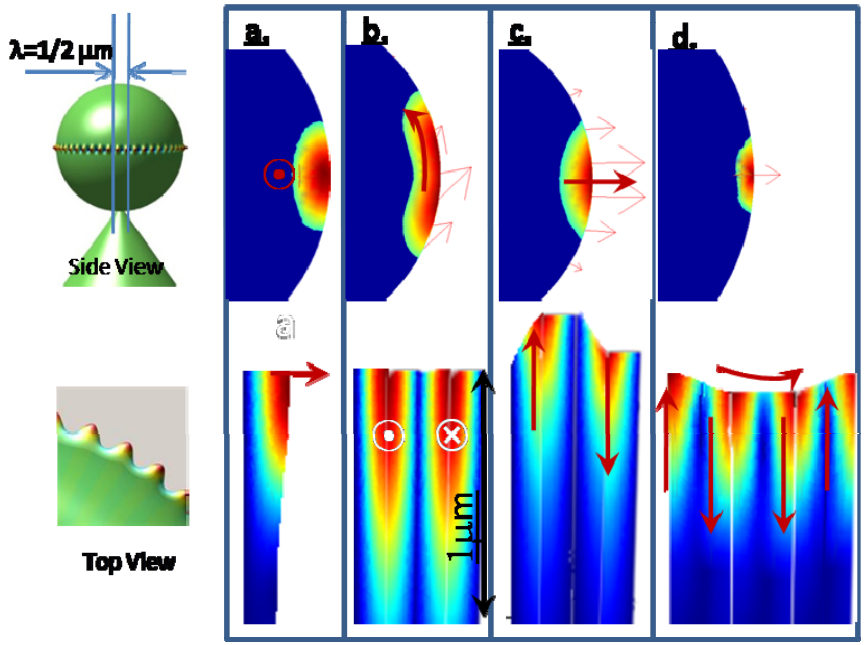


- **No notched support is needed, no lambda/4 isolators are needed**
  - Since sound is propagating azimuthally, always far from support
- **Can go high frequency irrespective of device miniaturization (fabrication limit)**
  - Since virtual electrodes are written with light (at an optical wavelength scale)

Conclusion

- Electrostrictive forces in optical MEMS should be able to excite transverse and Rayleigh waves in addition to longitudinal waves.
- Spherical boundary breaks symmetry to distinguish transverse polar and transverse radial waves.
- In each family, there exists higher order modes with multiple maxima along the radial and polar directions.
- A wide range of frequencies is achievable due to the different wave velocities.
- No notched support is needed as waves are propagating circumferentially

- Experimental results to follow soon.



Questions?